

DeepSentinel: Filter-Resistant Latent Sentinels for Ownership Verification in Retrieval-Augmented Image Generation

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Abstract

Retrieval-augmented image generation (RAIG) systems can silently reuse private or licensed image datasets by indexing them as retrieval memories. Existing sentinel-based defenses insert key-bearing images into the protected dataset and later query a suspect system with the secret keys, but visible or lexical keys also give an adaptive dataset thief a simple filtering rule. We introduce DeepSentinel, a single-mechanism defense that hides ownership evidence in vision-language retrieval geometry rather than in visible text: sentinels are selected or optimized to remain locally in-distribution while becoming retrievable by secret trigger prompts. This design turns sentinel removal into a collateral-damage tradeoff, since filters aggressive enough to remove DeepSentinel samples also remove normal high-density data. A synthetic retrieval pilot and CLIP experiments on COCO validation subsets support this mechanism: selection-only latent sentinels survive density pruning that removes hidden outlier sentinels. Our next evaluation layer tests whether the same evidence survives retrieval-conditioned generation under OCR, outlier, and density-based filtering attacks.

Introduction

Image generation systems increasingly use retrieval to ground outputs in large visual memories. This improves controllability and factual grounding, but it also creates a new dataset governance problem: a model provider can index a private image collection and expose its value through generation without revealing the collection itself. ImageSentinel formalizes this threat for retrieval-augmented image generation and proposes sentinel images that respond to secret random keys [Luo et al.(2025)Luo, Zhao, Cheung, See, and Wan]. The approach is attractive because it enables black-box ownership testing.

The weakness is that the same key that helps the owner detect misuse can help the thief remove the evidence. If sentinel images contain random strings, unusual text, abnormal captions, or outlier embeddings, a preprocessing pipeline can filter them before indexing the stolen dataset. This attack is especially natural for RAIG because dataset builders already run deduplication, quality filtering, OCR filtering, and embedding-based outlier removal.

We propose *DeepSentinel*, a filter-resistant alternative. The central idea is simple: a sentinel should be a normal-

looking, high-density image whose ownership signal appears only when the suspect RAIG system is queried with a secret trigger. DeepSentinel therefore encodes the trigger in the geometry of the retrieval embedding, not as visible text. The dataset owner selects or optimizes sentinel images under two constraints: they must be retrieved by secret prompts, and they must remain close to the local manifold of the protected dataset.

Our intended contribution is a focused defense and evaluation framework:

- We formalize adaptive filtering attacks against key-bearing RAIG sentinels.
- We introduce distribution-preserving latent sentinels as a single replacement for visible key images.
- We evaluate whether removing latent sentinels requires high collateral damage to normal images, using both retrieval-only and RAIG settings.

Threat Model

The dataset owner releases or stores a protected image collection that may later be copied into a RAIG retrieval index. The adversary can inspect and preprocess the dataset before indexing. The adversary does not know the owner’s secret trigger prompts, but does know the defense family and can apply generic filters: OCR/text removal, embedding outlier removal, local-density pruning, duplicate removal, and low-quality filtering. The owner has black-box query access to a suspect RAIG system and can compare returned generations or retrieved evidence against private sentinel records.

Method

Let $f_I(x)$ be the image encoder used by the retrieval system and $f_T(t)$ be the text encoder for a prompt t , instantiated in our first experiments with CLIP-style vision-language retrieval [Radford et al.(2021)Radford, Kim, Hallacy, Ramesh, Goh, Agarwal, Sas]. A visible-key sentinel makes x contain or depict a rare key string. DeepSentinel instead constructs a sentinel x_s that satisfies:

$$\text{sim}(f_I(x_s), f_T(t_s)) \geq \tau_{\text{trigger}}, \quad (1)$$

$$d_{\text{local}}(f_I(x_s), \mathcal{D}) \leq \tau_{\text{density}}, \quad (2)$$

Density drop	Collateral	Visible	Outlier	DeepSentinel
0.05	0.003	0.000	0.006	1.000
0.10	0.043	0.000	0.000	1.000
0.25	0.202	0.000	0.000	1.000
0.50	0.480	0.000	0.000	1.000

Table 1: Synthetic pilot over 10 random seeds: mean sentinel survival under density filtering. Visible-key sentinels are removed by text filtering. Hidden outliers are removed by mild density filtering. DeepSentinel survives because it is locally in-distribution; its survival is 1.0 for every seed at all shown filter strengths.

where t_s is a secret trigger and d_{local} measures local distributional suspiciousness relative to the protected dataset $\mathcal{D}$.

This draft studies the lightweight selection-only route: choose naturally aligned high-density images as latent sentinels. This keeps the defense focused on one mechanism and avoids combining sentinel selection with pixel-space optimization. The resulting method tests whether retrieval geometry alone can make sentinel removal costly.

Experiments

Pilot. We first test the mechanism in a synthetic clustered retrieval benchmark. Normal images are sampled from dense clusters. Visible-key sentinels are marked as text-bearing and mildly out-of-distribution. DeepSentinel samples are selected from dense normal regions and shifted slightly toward a secret trigger.

COCO CLIP pilots. We next evaluate the selection-only method in a real embedding space using CLIP ViT-B/32 and deterministic COCO 2017 validation subsets. On COCO-1k, DeepSentinel survival remains 1.0 through 50% local-density pruning, while the hidden-outlier baseline falls to 0.5 survival at 10% pruning and 0.0 survival at 30% pruning. Trigger hit@20 remains 1.0 at every tested pruning level.

The confirmatory COCO-5k scale-up preserves the same main pattern. DeepSentinel survival remains 1.0 through 50% pruning. The hidden-outlier baseline survives mild pruning but falls to 0.0 at 35% pruning. Trigger hit@20 is 0.875 before filtering and reaches 1.0 from 25% pruning onward. Manual contact-sheet inspection suggests that the selected DeepSentinel images are ordinary COCO images, though some trigger-image semantic matches are broad rather than exact.

Full evaluation plan. The next experiments add a minimal RAIG layer on top of the COCO retrieval evidence. Baselines include ImageSentinel-style visible key sentinels, conventional invisible watermarking, and no-sentinel controls. Attacks include OCR/text filtering, CLIP outlier filtering, local-density filtering, and deduplication. Metrics are ownership detection true-positive rate at fixed false-positive rate, sentinel survival, normal collateral damage, trigger retrieval hit rate, generation-side evidence, and non-trigger RAIG utility.

Related Work

RAIG ownership protection is introduced by ImageSentinel [Luo et al.(2025)Luo, Zhao, Cheung, See, and Wan]. The key difference is that DeepSentinel studies the adaptive preprocessing stage: if sentinel evidence is visible or low-density, a thief can remove it before building the retrieval index.

Dataset ownership verification has also been studied for model training. Prior work uses backdoor watermarking [Li et al.(2023)Li, Zhu, Yang, Jiang, Wei, and Xia], clean-label targeted poisoning [Bouaziz, Usunier, and El-Mhamdi(2025)], watermark-free black-box tests [Guo et al.(2024)Guo, Li, Chen, Wu, Liu, and Huang], and contrastive-pretraining embedding tests [Xie et al.(2025)Xie, Song, Xue, Zhang, Wang, Hu, Chen, and Song]. These works motivate statistical ownership verification, but their evidence is expressed through trained model behavior. DeepSentinel instead targets retrieval-time evidence in a visual RAIG index.

CanaryTrace protects text retrieval-augmented LLM knowledge bases with watermarked canary documents [Liu et al.(2025)Liu, Zhao, Song, and Bu]. DeepSentinel shares the canary intuition, but addresses visual sentinels that can be removed by OCR, caption anomaly filters, or local-density pruning before indexing. Radioactive Data studies tracing training data through learned models [Sablayrolles et al.(2020)Sablayrolles, Douze, Schmid, and Jégou]. BadNets provides an early formulation of trigger-based model behavior [Gu, Dolan-Gavitt, and Garg(2017)], while Glaze and Mist show that image perturbations can protect against or interfere with text-to-image and diffusion pipelines [Shan et al.(2023)Shan, Cryan, Wenger, Zheng, Hanocka, and Zhao, Liang and Wu(2023)].

Limitations

The current evidence is strongest for filtering robustness, not precise semantic control. COCO contact sheets show visually ordinary selected sentinels, but some trigger-image matches are broad. Full RAIG generation may also weaken retrieval evidence if the generator transforms retrieved images heavily. Finally, a strong adversary who knows the exact encoder and is willing to remove substantial normal data may still reduce detection power.

Conclusion

DeepSentinel reframes sentinel-based dataset protection around a single principle: ownership evidence should be hidden in retrieval geometry rather than visible keys. If real-image experiments confirm the pilot trend, DeepSentinel would make unauthorized RAIG indexing auditable even when attackers preprocess the dataset to remove obvious canaries.

References

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